

Homework 4

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13. Let X be the life of the component and $f_X(x)$ its density function. The desired (conditional) probability is

$$\begin{aligned} P(X < 11000 \mid X > 9000) &= \frac{P(9000 < X < 11000)}{P(X > 9000)} \\ &= \frac{\int_{9000}^{11000} f_X(x) dx}{\int_{9000}^{\infty} f_X(x) dx}. \end{aligned}$$

- (a) Assuming $X \sim \text{Exponential}(\lambda)$, where $\lambda = 1/10000$, since

$$f_X(x) = \lambda e^{-\lambda x} \cdot 1_{\{x>0\}},$$

the desired probability is

$$\begin{aligned} P(X < 11000 \mid X > 9000) &= \frac{\int_{9000}^{11000} f_X(x) dx}{\int_{9000}^{\infty} f_X(x) dx} \\ &= \frac{[1 - e^{\lambda x}]_{9000}^{11000}}{[1 - e^{\lambda x}]_{9000}^{\infty}} \\ &= \frac{(1 - e^{-1.1}) - (1 - e^{-0.9})}{e^{-0.9}} \\ &= \frac{e^{-0.9} - e^{-1.1}}{e^{-0.9}} \\ &= 1 - e^{-0.2}. \end{aligned}$$

- (b) Assuming $X \sim \text{Uniform}(8000, 12000)$, since

$$f_X(x) = \frac{1}{4000} \cdot 1_{\{x \in (8000, 12000)\}},$$

the desired probability is

$$P(X < 11000 \mid X > 9000) = \frac{(11000 - 9000)/4000}{(12000 - 9000)/4000} = 2/3.$$

16. Note that for $z > 0$ the CDF of $Z \sim \text{Exponential}(\lambda)$ is

$$P(Z < z) = 1 - e^{-\lambda z}$$

and its tail probability is

$$P(Z \geq z) = e^{-\lambda z}.$$

Let $M = \min(X, Y) > 0$. Given $m > 0$, by the independence assumption we can write $P((X \geq m) \cap (Y \geq m)) = P(X \geq m)P(Y \geq m)$ and hence

$$\begin{aligned} F_M(m) &= P(M < m) \\ &= P((X < m) \cup (Y < m)) \\ &= 1 - P((X \geq m) \cap (Y \geq m)) \\ &= 1 - P(X \geq m)P(Y \geq m) \\ &= 1 - e^{-\lambda_1 m} \cdot e^{-\lambda_2 m} \\ &= 1 - e^{-(\lambda_1 + \lambda_2)m}, \end{aligned}$$

which is the CDF of $\text{Exponential}(\lambda_1 + \lambda_2)$. By the uniqueness of CDF, we conclude that $\min(X, Y)$ has distribution $\text{Exponential}(\lambda_1 + \lambda_2)$.

22. Let $g(x) = 1/x$, which is convex on $(0, \infty)$. By Jensen's inequality we have

$$E(1/X) = E(g(X)) \geq g(E(X)) = 1/E(X).$$

23. (a) For any fixed $x \in [0, 1]$, we can write

$$f_{X,Y}(x, y) = \frac{1}{\pi} \cdot 1_{\{y^2 \leq 1-x^2\}} = \frac{1}{\pi} \cdot 1_{\{-\sqrt{1-x^2} \leq y \leq \sqrt{1-x^2}\}}.$$

So the marginal density function of X is

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy = \frac{1}{\pi} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} dy = \frac{2}{\pi} \sqrt{1-x^2}.$$

This is for $x \in [0, 1]$. For $x \notin [0, 1]$, $f_{X,Y}(x, y) = 0$ so $f_X(x) = 0$. More explicitly, the marginal density function of X is

$$f_X(x) = \begin{cases} \frac{2}{\pi} \sqrt{1-x^2} & \text{if } x \in [0, 1] \\ 0 & \text{otherwise} \end{cases}.$$

Similarly, the marginal density function of Y is

$$f_Y(y) = \begin{cases} \frac{2}{\pi} \sqrt{1-y^2} & \text{if } y \in [0, 1] \\ 0 & \text{otherwise} \end{cases}.$$

(b) For $0 < d \leq 1$, $F_D(d) = P(D \leq d)$ is the probability that a randomly selected point from the unit disk is in the circle with radius d , which is $d^2\pi/\pi = d^2$. For $d > 1$, $D \leq d$ always holds. So

$$F_D(d) = \begin{cases} 1 & \text{if } d > 1 \\ d^2 & \text{if } 0 < d \leq 1 \\ 0 & \text{if } d \leq 0 \end{cases}.$$

(c) Apply the identity established in Exercise 14

$$E(D) = \int_0^{\infty} P(D > x) dx.$$

Since the tail probability is $1 - F_D(x) = (1 - x^2) \cdot 1_{\{x \in [0, 1]\}}$, we have

$$E(D) = \int_0^1 (1 - x^2) dx = 2/3.$$

26. (a) For $t > 0$, we have

$$\begin{aligned}
P(Y < tX) &= \iint_{\{y < tx\}} f_{X,Y}(x,y) dx dy \\
&= \lambda_1 \lambda_2 \int_0^\infty \int_{y/t}^\infty e^{-\lambda_1 x} e^{-\lambda_2 y} dx dy \\
&= \lambda_2 \int_0^\infty e^{-\lambda_2 y} \left(\int_{y/t}^\infty \lambda_1 e^{-\lambda_1 x} dx \right) dy \\
&= \lambda_2 \int_0^\infty e^{-(\lambda_2 + \frac{\lambda_1}{t})y} dy \\
&= \frac{\lambda_2}{\lambda_2 + \frac{\lambda_1}{t}} \\
&= \frac{t\lambda_2}{\lambda_1 + t\lambda_2}.
\end{aligned}$$

(b) The CDF is

$$F_T(t) = P(T < t) = P(Y/X < t) = P(Y < tX) = \frac{t}{1+t}, \quad t > 0.$$

Thus for $t > 0$ the PDF is

$$f_T(t) = F'_T(t) = \frac{(1+t) - t}{(1+t)^2} = \frac{1}{(1+t)^2}.$$

Explicitly, the PDF is

$$f_T(t) = \begin{cases} \frac{1}{(1+t)^2} & \text{if } t > 0 \\ 0 & \text{otherwise} \end{cases}.$$

(c) Note that

$$E(T) = \int_0^\infty \frac{t}{(1+t)^2} dt.$$

As $t \rightarrow \infty$, the integrand decays to zero at the rate of $o(1/t)$ which is too slow for the integral to converge, so the $E(T) = \infty$. This is similar to the positive part of the Cauchy distribution.